India Semiconductor Market — Deep Research Brief for Equity Investors

Compiled: 14 June 2026. Multi-source, fact-checked. This is research and education, not investment advice. Verify all live prices, market caps, and P/E ratios against current exchange/Screener data before any decision.

TL;DR — The Investor's One-Page View

- **The consumption story is real; the manufacturing story is early.** India's chip *demand* is projected to roughly double to **~$100–120bn by 2030** (from ~$45–54bn in 2025), but India still **imports >90% of its chips** today and will keep importing most of them for years. Demand and domestic supply are on very different timelines.
- **India's genuine, durable edge is chip DESIGN**, not fabrication. ~20% of the world's chip-design engineers already sit in India (AMD, Nvidia, Qualcomm, Intel, Broadcom, Synopsys, Cadence captives). This is the lowest-risk, highest-moat exposure.
- **Manufacturing is currently a BACK-END (OSAT/ATMP) story** — assembly, test, packaging — which is lower-value and less moaty than front-end fabs. Three units are already in commercial/initial production (CG Power Aug 2025, Micron Feb 2026, Kaynes Mar 2026).
- **The marquee front-end fab (Tata–PSMC, Dholera, ₹91,000 cr, 28nm) is still pre-revenue.** "First silicon" is targeted **late/Dec 2026** but has a real risk of slipping to 2027. This is the single biggest sentiment catalyst/risk for the theme.
- **Listed-equity froth is a live concern.** The premium OSAT/design names (Kaynes ~98x trailing, CG Power ~80x+, Tata Elxsi de-rated from ~60x) had a *tough FY26* even as the narrative stayed bullish. The biggest fab exposure (Tata Electronics) is **not investable** via listed equity.
- **Even NITI Aayog is steering India AWAY from leading-edge fabs**, toward advanced packaging / compound semiconductors ("More-than-Moore"). Underwrite execution risk heavily; the failed-deal history (Vedanta-Foxconn, ISMC) is a credibility overhang.

1. Market Size & Growth — Read Every Number With Its Scope

Estimates vary widely because firms measure different things: **end-demand consumption** vs. **domestic production/localization** vs. **total addressable market**. This is the single biggest area of source disagreement.

Metric	Figure	Source	Date
End-demand (consumption)	**$54bn (2025) → $108bn (2030)**, ~15% CAGR	UBS (via IBEF, Free Press Journal)	Apr 2025
Localization revenue by 2030	$13bn (India ~6.5% of global demand)	UBS	Apr 2025
Market by 2030 / 2035	**$120bn (2030); $300bn (2035)**, ~20% CAGR	Deloitte "TMT Predictions 2026"	Mar 2026

Metric	Figure	Source	Date
Current size + imports	**$45–50bn (FY25); >90% imported**; local to meet >60% by 2035	Deloitte	Mar 2026
Capex pipeline	+$50bn (2025–30), +$75–80bn (2030–35); ~2M jobs by 2035	Deloitte	Mar 2026
Long-run value chain	**$120–150bn by 2035** (top-6 chip nation by 2032)	NITI Aayog roadmap	May 2026
Cumulative imports FY17–FY25	**~$150bn** spent on chip imports	NITI Aayog	May 2026

Conflict flagged: 2025 base ranges $45bn (Deloitte) → $54bn (UBS) → ~$60bn (IMARC). 2030 target ranges $100–120bn. CAGR ranges 12.5%–20%. The **$108bn (UBS) and $120bn (Deloitte) for 2030** are the most-cited; Deloitte's 20% CAGR / $300bn-by-2035 is the **bull-case outlier**. Use ranges, not point estimates.

Investor read: The entire thesis is **import substitution + localization**. Demand drivers are genuine — smartphones (India = #2 global mobile manufacturer), EVs/automotive, telecom/5G, data centers, AI/IoT.

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2. The Policy Stack — Top-Down Catalysts to Watch

Scheme	Outlay	Status / Notes
ISM 1.0 (India Semiconductor Mission)	**₹76,000 cr (~$9–10bn)**, up to 50% capex support	Approved Dec 2021; funded the current project pipeline
ISM 2.0	**₹1,000 cr FY27 budget line** to launch; reported **₹40,000 cr** headline; proposed **₹1–1.2 lakh cr** total *pending final Finance Ministry/Cabinet approval*	Launched in **Union Budget FY2026-27 (Feb 2026)**. Pivots from "fabs only" → **equipment, materials, full-stack Indian IP, R&D, supply-chain resilience**
DLI (Design Linked Incentive)	Up to 50% of design cost (cap ₹15 cr) + 4–6% of net sales (cap ₹30 cr)	**23 design projects approved**; 72 firms with EDA-tool access via C-DAC. Targets 100 design companies
ECMS (Electronics Components Mfg Scheme)	**₹22,919 cr (~$2.7bn)**, notified Apr 2025	The big upstream/components complement to ISM. Already drew **₹54,567 cr** → **₹1.15 lakh cr** in commitments (~2x target) across 46+ approvals
Modified Semicon/ Display Programme	**₹8,000 cr** for FY2026-27	Active

Critical nuance — don't conflate two "ISM 2.0 budget" numbers: - **₹1,000 cr** = actual FY2026-27 budget line item to *launch* ISM 2.0. - **₹40,000 cr / ₹1–1.2 lakh cr** = reported total program outlay still under inter-ministerial consultation, **not yet formally cabinet-approved at that quantum** as of the latest sources. - **Watch this approval as a key catalyst.** Also note: ISM 2.0 came in the **FY2026-27 budget (Feb 2026)**, not FY2025-26.

ISM project pipeline: ~12–13 projects, ~₹1.6–1.64 lakh crore committed across 6–7 states (a moving target: 10 in late 2025 → 12–13 by May 2026). Gujarat leads (4 projects, 3 in Sanand + the Dholera fab); UP offers the most generous headline incentive (ceiling up to 100% of eligible cost); Assam hosts Tata's OSAT.

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3. The Manufacturing Pipeline — Mostly Back-End Today

Headline: India is currently an **OSAT/ATMP + compound/advanced-packaging play, NOT yet leading-edge logic.** Of ~10–13 approved units, only **one is a large logic fab**.

The one front-end fab

Tata Electronics + PSMC (Powerchip, Taiwan) — Dholera, Gujarat - **~₹91,000 cr (~$11bn)**; **28nm** initially (range 28–110nm: analog/logic, PMICs, display drivers, MCUs, automotive). NOT 3nm/2nm. - Capacity **50,000 wafers/month** (~3bn chips/yr). **ASML lithography pact** signed ~16 May 2026. - Construction ~50% complete; redesigned for Dholera soil conditions. **First silicon targeted late/Dec 2026** — *with real slippage risk to 2027.*

Back-end (OSAT/ATMP) units already producing or near

Unit	Investor	Location	Investment	Status (2026)
Micron ATMP	Micron (US)	Sanand, GJ	**~₹22,500 cr ($2.75bn)**	**Inaugurated 28 Feb 2026**; first Made-in-India memory modules shipping
CG Semi OSAT	CG Power (92.3%) + Renesas + Stars	Sanand, GJ	**₹7,600 cr**	Inaugurated Aug 2025; first chip from pilot mid-2026
Kaynes Semicon OSAT	Kaynes Technology	Sanand, GJ	**₹3,307 cr**	**Shipped India's first commercial multi-chip module (Oct 2025)**; PM inaugurated 31 Mar 2026
TSAT (Tata) OSAT	Tata Electronics	Jagiroad, Assam	**~₹27,000 cr**	Phase-1 commissioning ~Apr 2026; up to 48M chips/day
HCL–Foxconn JV	HCL + Foxconn	Jewar, UP	**₹3,706 cr**	Cabinet-approved May 2025; display-driver ICs; **operational ~2027–28**

Newest units (~Aug 2025, ₹4,600 cr combined)

- **SiCSem** (Odisha) — India's first commercial **compound (SiC) fab** + packaging
- **3D Glass Solutions** (US, Odisha) — embedded glass substrate / advanced packaging
- **CDIL / Continental Device** (Punjab) — power discretes (MOSFETs, IGBTs, Si + SiC)
- **ASIP Technologies** (w/ APACT, Andhra Pradesh) — advanced packaging / 3DHI
- Plus newer May 2026 approvals: **Crystal Matrix** (Micro-LED/GaN, Dholera), **Suchi Semicon** (OSAT, Surat).

"Who made India's first chip?" is contested: Kaynes (first commercial *MCM shipment*, Oct 2025) vs CG Power (first to *inaugurate/pilot*, Aug 2025) vs Micron (first *large ATMP at scale*, Feb 2026). Different outlets crown different "firsts" by definition.

Timeline slippage is the norm: Micron (2025→Feb 2026); Tata Assam (mid-2025→Apr 2026); Dholera fab (mid-2025→late 2026, possibly 2027).

4. Listed Indian Stocks With Semiconductor Exposure

> **No buy/sell advice.** Exposure and analyst-cited views only. Figures from aggregator snippets — **verify before acting.** "Semiconductor stock" lists conflate very different exposures (true OSAT vs design services vs power-semi vs EMS vs pure theme tags).

True OSAT / manufacturing

- **CG Power & Industrial Solutions (CGPOWER)** — "CG Semi" OSAT JV (92.3%) with Renesas. Commercial production imminent (Mar 2026). **Valuation rich: ~80x+ 2026 P/E** vs industry ~66x; Nuvama cautious.
- **Kaynes Technology (KAYNES)** — integrated ESDM + OSAT (Sanand, inaugurated Mar 2026) + HDI PCB. FY26 revenue ₹3,626 cr (+33%). **Stock fell ~27%** on weak Q4 FY26. **Very high valuation: trailing P/E ~98x** (some snapshots higher); Nuvama HOLD — "valuations yet to become attractive." OSAT revenue still tiny (<₹1bn FY26).
- **HCLTech (HCLTECH)** — via HCL–Foxconn OSAT JV (Jewar, display-driver ICs), but described as a *mid-tier OSAT, not a fab*; operational ~2027–28. Exposure is small relative to HCLTech's core IT business.

Design / engineering services ("picks & shovels")

- **Tata Elxsi (TATAELXSI)** — closest listed design-services proxy; asset-light, high-margin. But **FY26 growth stalled** (revenue +0.8%, net income −20%), stock **down ~27–32% YoY**; de-rated from ~60x to ~35–47x. Multiple Feb 2026 downgrades (Kotak cut target ₹7,000→₹5,500).
- **Tata Technologies (TATATECH)** — ER&D, auto-heavy, peripheral design exposure; downgraded by some on valuation.
- **MosChip (532407)**, **Cyient (CYIENT)** (semicon unit still loss-making, breakeven ~FY27), **L&T Technology Services (LTTS)**, **ASM Technologies (ASMTEC)** — smaller design/services names.

Power-semi / compound

- **RIR Power Electronics (517035)** — power semis; building India's first SiC plant (Odisha, ~₹618 cr); SiC epi-wafer production ~Q2 FY27. FY26 PAT −18.8%.

EMS / components (adjacent, not chip fab)

- **Dixon Technologies (DIXON)** — largest Indian EMS; pushing into components/displays (Dixon Display Technologies got ECMS approval Mar 2026, HKC JV). Market cap ~₹70,000 cr.
- **Cyient DLM (CYIENTDLM)** — EMS/box-build; DLM segment revenue **−~30% YoY** (pushouts, tariffs); mostly HOLD.
- **Netweb Technologies (NETWEB)** — AI servers / high-end computing (system integration, not chips); FY26 revenue +92%.

Stalled / not investable

- **Tata Electronics** (the actual fab + OSAT) — **UNLISTED**; only indirect via unlisted Tata Sons. The biggest fab exposure has no clean listed vehicle.

- **Vedanta (VEDL)** — semicon story **effectively stalled** since Foxconn's 2023 exit; live asset is display glass (AvanStrate), not chips.
- **SPEL Semiconductor (517166)** — legacy OSAT, tiny/loss-making. **MIC Electronics (539771)** — loose "theme" tag, thin real exposure. **Polymatech** — unlisted.

Funds / ETFs

- **No dedicated onshore (domestic) semiconductor mutual fund/ETF exists** in India as of 2026. Lists labeled "semiconductor funds" are usually diversified tech funds (Motilal Oswal Flexi Cap, Tata Digital India) — *not* pure semicon.
- **Mirae Asset Global Allocation Fund (GIFT City IFSC)** — launched Apr 2025; Category III AIF, **non-retail** (accredited investors), globally diversified incl. AI/semiconductor themes. Resident access via LRS.
- Indirect global semicon exposure via Nasdaq/US-tech FoFs (Motilal Oswal, Edelweiss).

Valuation / froth warning

Both Kaynes and CG Power flagged by Nuvama as too expensive for entry; several "theme" names (MIC, SPEL, parts of the basket) carry thin or loss-making *actual* semiconductor revenue — classic theme-froth. Global backdrop is also stretched (PHLX Semiconductor Index ~53x P/E vs ~28x 10-yr avg). **Optimists** cite ISM execution; **skeptics** note OSAT is the low-margin, low-tech end and most Indian OSAT revenue is still nascent.

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5. Global Players & Supply-Chain Positioning

India's structural moat = design. ~20% of the world's chip-design engineers are in India (Bastion Research, Sep 2025). A 3nm chip was *designed* (not fabricated) in India (PIB, 2025).

Global firm	India footprint
AMD	Largest global design center, Bengaluru; ~3,000 engineers + 3,000 more by 2028; $400M/5yr
Applied Materials	Bengaluru engineering center; ~$400M/4yr, supports >$2bn investment
Lam Research	**~$1.2bn Karnataka commitment** — R&D lab + silicon-component plant; opened India Center for Engineering
Micron	$2.75bn ATMP, Sanand (operating since Feb 2026)
Nvidia	India Deep Tech Alliance ($2bn pledge w/ Qualcomm); Blackwell GPUs via Yotta/E2E; custom-AI-chip talks
Analog Devices	Strategic MoU with Tata (make ADI products at Dholera/Assam)
ASML	Lithography pact with Tata Electronics (~May 2026)
Synopsys / Cadence	Massive India R&D; Synopsys India ~19,800 employees; EDA tools seeded in 315 institutions
Also: **NXP** (4 design centers, auto AI), Intel, Qualcomm, TI, Marvell, Infineon, Renesas	

Supply-chain segments — where India stands

Segment	India position
Design	**STRONG** — talent + MNC captives (lowest-risk exposure)
EDA	**STRONG** (as contributor/consumer)
ATMP/OSAT	**EMERGING** — the entry point; market ~$1.7bn (2025) → ~$3bn (2032)
Fabrication	**NASCENT** — first fab trial late 2026; no HVM yet
Equipment	**DEVELOPING** — Lam/Applied anchoring; ~$4.9bn by 2032
Materials & gases	**WEAK GAP** — >90% imported; ISM 2.0 explicitly targets this (e.g., INOX Air Products specialty-gas hub at Dholera)

Geopolitics — the China+1 swing factor

China imposed export curbs on **gallium, germanium, antimony, rare earths** (Dec 2024 → expanded Oct 2025), then **suspended them (Nov 2025)** under a ~1-year US-China truce. **Investor read:** these elements matter directly to SiC/compound-semi and power/RF chips, and Chinese dominance is *why* Western OEMs court India's "trust supply chain." But the Nov 2025 truce **eased near-term pressure** — so China+1 is a **structural/multi-year thesis, not an acute 2026 squeeze.** Watch the **1-year truce expiry** as the next catalyst/risk.

Talent: ISM targets training 85,000 engineers (10-yr); ~67,000 trained as of Feb 2026; EDA tools in 315 institutions (→500 under ISM 2.0).

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6. Risks & Headwinds — Underwrite These Heavily

1. **Import dependence:** >90% of chips and >90% of fab inputs (high-purity chemicals, specialty gases, silicon wafers) imported. ~$150bn spent on imports FY17–FY25.
2. **Capital intensity:** NITI estimates **$135–180bn** cumulative investment needed over the decade.
3. **Infrastructure:** Fabs need ultra-pure water (parts-per-billion) and uninterrupted power — both flagged as risks even at Dholera/YEIDA.
4. **Talent gap on the *manufacturing* side** (design is strong; fab/process talent is scarce).
5. **Subsidy-scale mismatch:** US CHIPS Act $52.7bn; China ~$150bn; Korea/Taiwan decades of backing. **Arizona alone will out-build India's planned capacity on more advanced nodes.** India's first fabs are mature (28nm+).
6. **Milestone-linked subsidy risk:** Disbursements tied to production milestones — a single contamination event during tool qualification can delay first silicon by months, with direct financial consequences.
7. **Execution track record: Vedanta-Foxconn JV collapsed (Jul 2023, ~$19.5bn)**; **ISMC/Tower (Karnataka, $3bn) stalled** amid legal disputes. Credibility overhang.
8. **Rare-earth supply shock** could delay Tata's rollout (raised internally, early 2026).
9. **Cyclicality:** global semiconductor oversupply/downturn cycles.

Notable strategic signal: NITI Aayog (May 2026) explicitly recommends India **NOT chase leading-edge wafer fabs** and instead pursue **"More-than-Moore"** — advanced packaging, system integration, compound semiconductors — targeting a top-3 global slot in advanced packaging by 2035. That's an institutional acknowledgment of the bear case on cutting-edge fabs.

7. Timeline Realism

- **Now (2025–26):** Back-end (OSAT/ATMP) operational — Micron, Kaynes, CG Power, Bhiwadi. *Packaging, not fabrication.*
- **Late/Dec 2026:** Tata-PSMC Dholera "first silicon" targeted — **plausible slip to 2027** (Digitimes reports fresh delays, leadership churn, rare-earth risk).
- **2027–28:** Realistic window for *meaningful, scaled* domestic **fab** output (mature nodes only). Tata capacity expected to "stabilize and scale."
- **2032 / 2035:** India targets top-6 chip nation (2032) and a $120–150bn value chain (2035) per NITI.

8. What to Watch (Catalysts & Checkpoints)

1. **Dholera "first silicon" (target Dec 2026)** — the single biggest sentiment catalyst/risk for the theme.
2. **Formal cabinet approval of ISM 2.0's full ₹40,000 cr / ₹1–1.2 lakh cr outlay** (currently only a ₹1,000 cr FY27 line).
3. **OSAT ramp curves** at Kaynes, CG Power, Micron — revenue actually showing up vs. slideware.
4. **US-China truce expiry (~late 2026)** — could re-ignite the China+1 / critical-minerals thesis.
5. **Valuation re-rating** — whether premium names (Kaynes, CG Power) grow into multiples or de-rate further (as Tata Elxsi did in FY26).
6. **Latest PIB releases** for exact, current ISM project count and committed capital.

How to Think About Exposure (Framework, Not Advice)

Risk appetite	Where the research points
Lower-risk / proven moat	**Design & engineering services** (talent arbitrage, asset-light) — but watch the FY26 growth slowdown that hit Tata Elxsi
Direct manufacturing exposure	**OSAT/ATMP** names (CG Power, Kaynes) — real production now, but **rich valuations** and low-margin end of the chain
The fab "moonshot"	**Largely uninvestable** via listed equity (Tata Electronics unlisted); the public proxy is indirect
Picks & shovels	EMS/components (Dixon, Cyient DLM), power-semi (RIR), AI servers (Netweb) — adjacent, varied quality
Diversified / global	No onshore pure-play fund; offshore (Mirae IFSC, non-retail) or global tech FoFs

Three Things to Verify Before Acting on Any Figure

1. **Latest ISM project count & committed capital** — confirm against the current PIB release (the number moved 10→13 in ~6 months).

2. **Whether ISM 2.0's full outlay has formal Cabinet approval** — only the ₹1,000 cr FY27 seed is confirmed.
3. **The scope basis of any market-size number** (consumption vs. local production) — this drives the ~2x spread in 2030 estimates.

Source Notes & Reliability

- **Most reliable / primary:** PIB / MeitY / NITI Aayog releases, company press releases (Tata, Micron, Renesas, HCL, AMD, Lam, ASML), Reuters-linked Evertiq, Business Standard, Economic Times, Mint, TechCrunch, EE Times, Outlook Business, Renesas/Business Wire.
- **Research houses (directional, methodology varies):** UBS, Deloitte, IMARC, Mordor, Counterpoint, PS Market Research, Bastion Research.
- **Caveats:** Several PIB/premium pages returned HTTP 403 to automated fetch; those figures were corroborated via reputable secondary outlets and are flagged inline. **Live equity prices/market caps/P-E ratios are from aggregator snapshots and may be stale — verify against Screener/exchange data.** The **Tata-ASML pact (May 2026)**, while well-reported, was not confirmed against a tier-1 wire or ASML/Tata primary release at compile time.

Methodology: five parallel research streams (market/policy, projects, listed stocks, global players/supply chain, risks/recent developments), each running 8–12 web searches and source fetches, with cross-checking and explicit flagging of source disagreements. Not investment advice — do your own due diligence.